Lecture 1: Understanding Biochemistry and Biomolecules

1. Biochemistry

Biochemistry is the branch of science that studies the chemical processes and substances inside living organisms.

Biomolecules are substances produced by cells and living organisms, include carbohydrates, lipids, nucleic acids, and proteins.

2. Water and Its Properties

Water is the most important molecule in living cells.

Key properties of water:

- **Universal solvent** – dissolves many substances (like salt, sugar)
- **Temperature regulator** – helps control body temperature
- **High heat capacity** – absorbs heat without changing temperature quickly
- **Cohesion & adhesion** – helps water move in plants
- **Medium for reactions** – most biochemical reactions happen in water

Key Properties of Water

Polarity: Oxygen attracts electrons more strongly, creating slight negative charge on oxygen and positive on hydrogens, leading to polar covalent bonds and attraction between molecules.

Hydrogen Bonding: Strong attraction between the slightly positive hydrogen of one molecule and the slightly negative oxygen of another, causing many unique properties.

Universal Solvent: Its polarity allows it to dissolve many ionic and polar substances (like salts, sugars) by disrupting their bonds.

Cohesion & Adhesion: Cohesion (water sticks to water) creates high surface tension; adhesion (water sticks to other surfaces) enables capillary action, crucial for plants.

High Heat Capacity: Takes significant energy to change water's temperature, stabilizing temperatures in organisms and environments.

High Heat of Vaporization: Lots of energy needed to turn liquid water into steam, aiding in cooling (sweating).

Density Anomaly: Solid ice is less dense than liquid water (it floats) because its hydrogen bonds form a crystal lattice, preventing molecules from packing as closely as in liquid form.

Amphoteric Nature: Can act as both an acid (donating a proton) and a base (accepting a proton).

Why important?

Without water, cells cannot survive or function.

3. Biological Functions of Biomolecules

Carbohydrates (carbs) are a type of macronutrient found in certain foods and drinks. Sugars, starches and fiber are carbohydrates.

- Provide or Main source of energy
- Examples: glucose, starch, rice, bread
- Used by cells to make ATP (energy)

2. Proteins refers to the macronutrient found in many foods and drinks.

Made of amino acids

What is the function of proteins?

Proteins keep everything in your body working and moving so you can keep moving. Proteins are so important that scientists describe them as your body's workhorses.

Here are some things proteins do for your body:

- Help cells in your body "read" your genes and use that information to build new cells and molecules, including new proteins (yes, that's right: Proteins indeed help create other proteins)
- Provide structure to cells and tissues throughout your body
- Attach to invaders, like bacteria and viruses, to clear them from your body
- Break down nutrients in the foods you eat — including fats, proteins and carbohydrates — into forms your body can use (this might be another surprise: Some proteins, called enzymes, break down other proteins!)
- Send chemical signals and carry information from cell to cell
- Speed up (catalyze) chemical reactions within cells, which helps create the energy your body needs to function
- Serve as channels (like little doorways) in a cell's outer membrane that allow substances to pass in or out of the cell

- Transport small molecules throughout your body (for example, a protein in your blood called hemoglobin is like a special parcel service that carries oxygen to all your body's tissues)

Examples of proteins

Proteins go by different names depending on what they do. They also fall into some general categories according to their function. Here are some examples:

- **Defense proteins** protect your body. If you've heard of antibodies, then you're already familiar with one key type of defense protein. Antibodies are the hard workers that help your body ward off germs and other invaders.
- **Enzymes** serve as catalysts for chemical reactions, like in digestion. Enzymes make chemical reactions go at least 1 million times faster. Enzymes include lipase, which helps break down dietary fats, and carbohydrase, which breaks down carbs. Other enzyme examples include trypsin, thrombin and pepsin.
- **Messenger proteins** send chemical signals between different cells, tissues and organs in your body. Certain hormones are classified as messenger proteins because of the work they do to support body processes.
- **Motor proteins** help things move within and between your cells. For example, they support the movement of cell parts that are vital for wound healing. Motor proteins also help your muscles move, help food move through your digestive tract and support blood flow through your body.
- **Structural proteins** offer support for your body's cells. These proteins also help various cells and substances move within your body. Examples of structural proteins include collagen, tubulin and actin. Tubulins join up to form tubes that carry substances within your cells. Actin works with motor proteins to help your muscles contract.
- **Transport/channel proteins** act as little doorways for molecules to move between parts of a cell or between a cell and its surrounding environment. These proteins are often embedded within a cell's protective outer barrier (cell membrane).

3. **Lipids** are chemical compound or elements that they bond chemically to make lipids and lipids is made of carbon hydrogen and oxygen

What do lipids do?

Various lipids have different functions. These include:

- Triglycerides transport and store energy.
- Steroid hormones send messages in cells.
- Bile salts from cholesterol help digest fat.
- Fatty acids metabolize to create energy.
- Phospholipids and cholesterol make biological membranes. Phospholipids are like gatekeepers that let some molecules into your cells but require others to come in with proteins.

4. **Nucleic acids** are naturally occurring chemical compounds that serve as the primary information-carrying molecules in cells. They play an especially important role in directing protein synthesis. The two main classes of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).

Functions:

- Store genetic information
- Control protein production
- Pass traits from parents to offspring

Function: Genetic information storage

What is a Prokaryotic Cell?

Prokaryotic cells are organisms whose cells lack a nucleus and other organelles
Prokaryotes include Bacteria and Archaea.
The photosynthetic prokaryotes include cyanobacteria that perform photosynthesis.
A prokaryotic cell consists of a single membrane and therefore, all the reactions occur within the cytoplasm. They can be free-living or parasites.

Components of Prokaryotic Cells

The prokaryotic cells have four main components:

Plasma Membrane- It is an outer protective covering of phospholipid molecules which separates the cell from the surrounding environment.

Cytoplasm- It is a jelly-like substance present inside the cell. All the cell organelles are suspended in it.

DNA- It is the genetic material of the cell. All the prokaryotes possess a circular DNA. It directs what proteins the cell creates. It also regulates the actions of the cell.

Ribosomes- Protein synthesis occurs here.

Some prokaryotic cells possess cilia and flagella which helps in locomotion.

What is a Eukaryotic Cell?

eukaryote, any cell or organism that possesses a clearly defined nucleus. The eukaryotic cell has a nuclear membrane that surrounds the nucleus, in which the well-defined chromosomes (bodies containing the hereditary material) are located.

Characteristics of Eukaryotic Cells

The features of eukaryotic cells are as follows:

- Eukaryotic cells have the nucleus enclosed within the nuclear membrane.
- The cell has mitochondria.
- Flagella and cilia are the locomotory organs in a eukaryotic cell.
- A cell wall is the outermost layer of the eukaryotic cells.
- The cells divide by a process called mitosis.
- The eukaryotic cells contain a cytoskeletal structure.
- The nucleus contains a single, linear DNA, which carries all the genetic information.

Main parts:

Nucleus, Mitochondria, Endoplasmic reticulum, Golgi apparatus, Lysosomes, (Plants only): chloroplasts, cell wall, large vacuole.

Similarities between eukaryotic and prokaryotic cells

Cell Membrane

Both eukaryotic and prokaryotic cells bear a lipid bilayer, which is an arrangement of phospholipids and proteins that acts as a selective barrier between the internal and external environment of the cell.

Genetic Material

Eukaryotic and prokaryotic cells both use deoxyribonucleic acid (DNA) as the basis for their genetic information. This genetic material is needed to regulate and inform cell function through the creation of RNA by transcription, followed by the generation of proteins through translation.

Ribosomes

Ribosomes facilitate RNA translation and the creation of protein, which is essential to the functioning of both eukaryotic and prokaryotic cells.

Cytoplasm

The cytoplasm is the medium in which the biochemical reactions of the cell take place, of which the primary component is cytosol.

In eukaryotic cells, the cytoplasm comprises everything between the plasma membrane and the nuclear envelope, including the organelles; the material within the nucleus is termed the nucleoplasm. In prokaryotes the cytoplasm encompasses everything within the plasma membrane, including the cytoskeleton and genetic material.

Differences between eukaryotic and prokaryotic cells

Cell size

Eukaryotic cells are ordinarily larger (10 – 100um) than prokaryotic cells (1 – 10um).

Cell arrangement

Eukaryotes are often multicellular whereas prokaryotes are unicellular. There are however some exceptions –unicellular eukaryotes include amoebas, paramecium, yeast.

True membrane-bound nucleus

Eukaryotic cells have a true nucleus bound by a double membrane. It contains the DNA-related functions of the large cell in a smaller enclosure to ensure close proximity of materials and increased efficiency for cellular communication and functions.

In contrast, the smaller prokaryotic cells have no nucleus. The materials are already fairly close to each other and there is only a "nucleoid" which is the central open region of the cell where the DNA is located.

DNA structure

Eukaryotic DNA is linear and complexed with packaging proteins called "histones," before organization into a number of chromosomes

Prokaryotic DNA is circular and is neither associated with histones nor organized into chromosomes. A prokaryotic cell is simpler and requires far fewer genes to function than the

eukaryotic cell. Therefore, it contains only one circular DNA molecule and various smaller DNA circlets (plasmids).

Membrane-bound organelles

Eukaryotic cells contain many membrane-enclosed, large, complex organelles in the cytoplasm whereas prokaryotic cells do not contain these membrane-bound organelles.

This is a key difference because it allows a high level of intracellular division of labor and contributes to the greater complexity characteristic of eukaryotic cells.

Due to the larger size of the eukaryotic cells, confining certain cellular process to a smaller area also increases the efficiency of functions by improving communication and movement within the cell.

Only eukaryotes possess a membrane-bound nucleus and membrane-bound organelles such as the mitochondria, golgi apparatus, lysosomes, peroxisomes and ER.

Ribosome size

Both eukaryotic and prokaryotic cells contain many ribosomes; however the ribosomes of the eukaryotic cells are larger than prokaryotic ribosomes i.e. 80S compared to 70S.

Eukaryotic ribosomes also show more complexity than prokaryotic – they are constructed of five kinds of ribosomal RNA and about eighty kinds of proteins. In contrast, prokaryotic ribosomes are composed of only three kinds of rRNA and about fifty kinds of protein.

Cytoskeleton

This is a multicomponent system in eukaryotes composed of microtubules, actin filaments and intermediate filaments. It is required for maintaining cell shape, providing internal organization and mechanical support. It is also paramount in movement and cell division.

Sexual reproduction

Most eukaryotes undergo sexual reproduction whilst prokaryotes reproduce asexually. Sexual reproduction in eukaryotes results in offspring with genetic material which is a mixture of the parents' genome and during this process, genetic variation is generated via sexual recombination.

On the other hand, a prokaryote will reproduce clones of itself via binary fission and relies more on horizontal genetic transfer for variation.

Cell division

This occurs by mitosis for eukaryotic cells and binary fission for prokaryotic cells.

Eukaryotic cells undergo mitosis then cytokinesis. This involves numerous stages - the nuclear membrane disintegrates then the chromosomes are sorted and separated to ensure that each daughter cell receives two sets (a diploid number) of chromosomes. Following this, the cytoplasm divides to form two genetically identical daughter cells i.e. cytokinesis.

In contrast, prokaryotes undergo a simpler process of binary fission. This is faster than mitosis and involves DNA (nucleoid) replication, chromosomal segregation, and ultimately cell separation into two daughter cells genetically identical to the parent cell. Unlike mitosis, this process does not involve the nuclear envelope and centromere and spindle formation.